

Automated Test Report

26-12-2025

DUT Name	BBNL MPPT 1kW – [MPPT DC/DC Converter]
DUT Part number	TPS1030004A
DUT Rev & Checksum	S04 – FFABD
Testing person	VISHNU C V

Test Report Summary

SI No	Test Name	Result
1	Input voltage range verification	Fail
2	Output voltage accuracy	Fail
3	Load regulation	Pass



PowerSafe-X

Test Item 1: Input voltage range verification

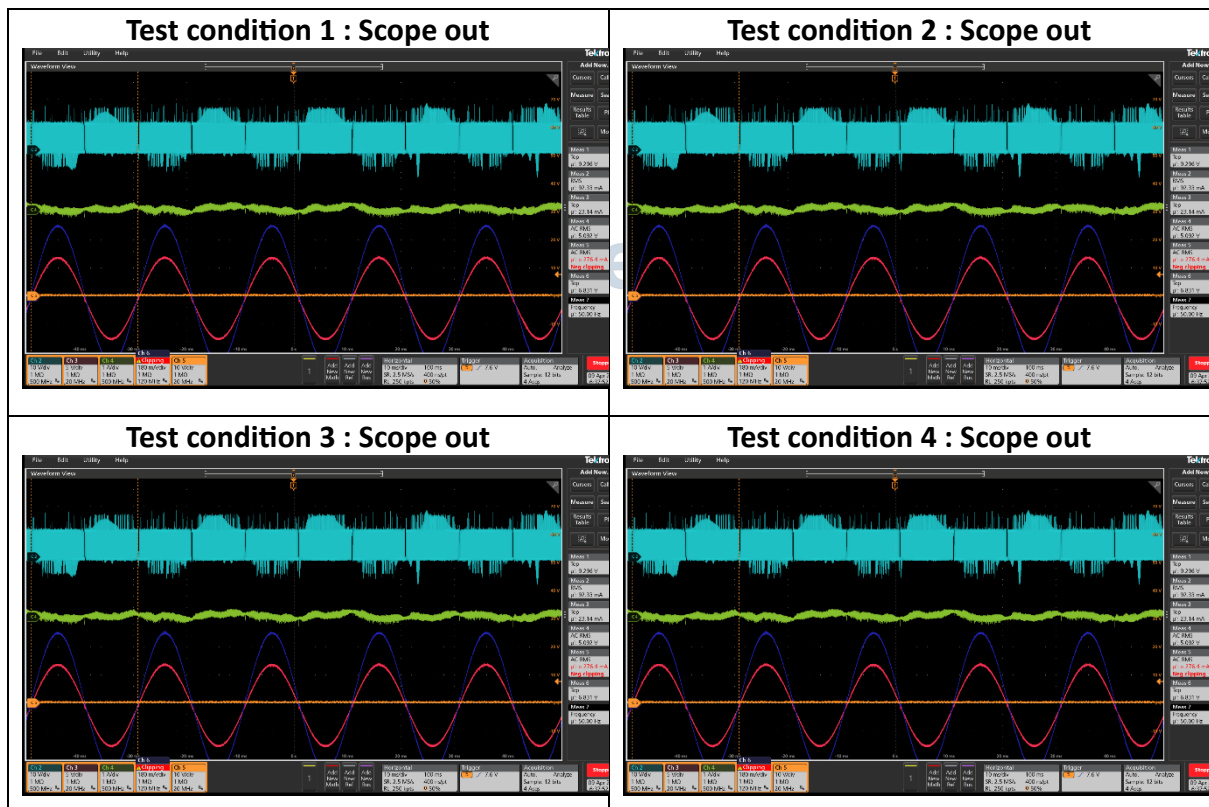
Verification of Input voltage range verification

Test Spec:

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



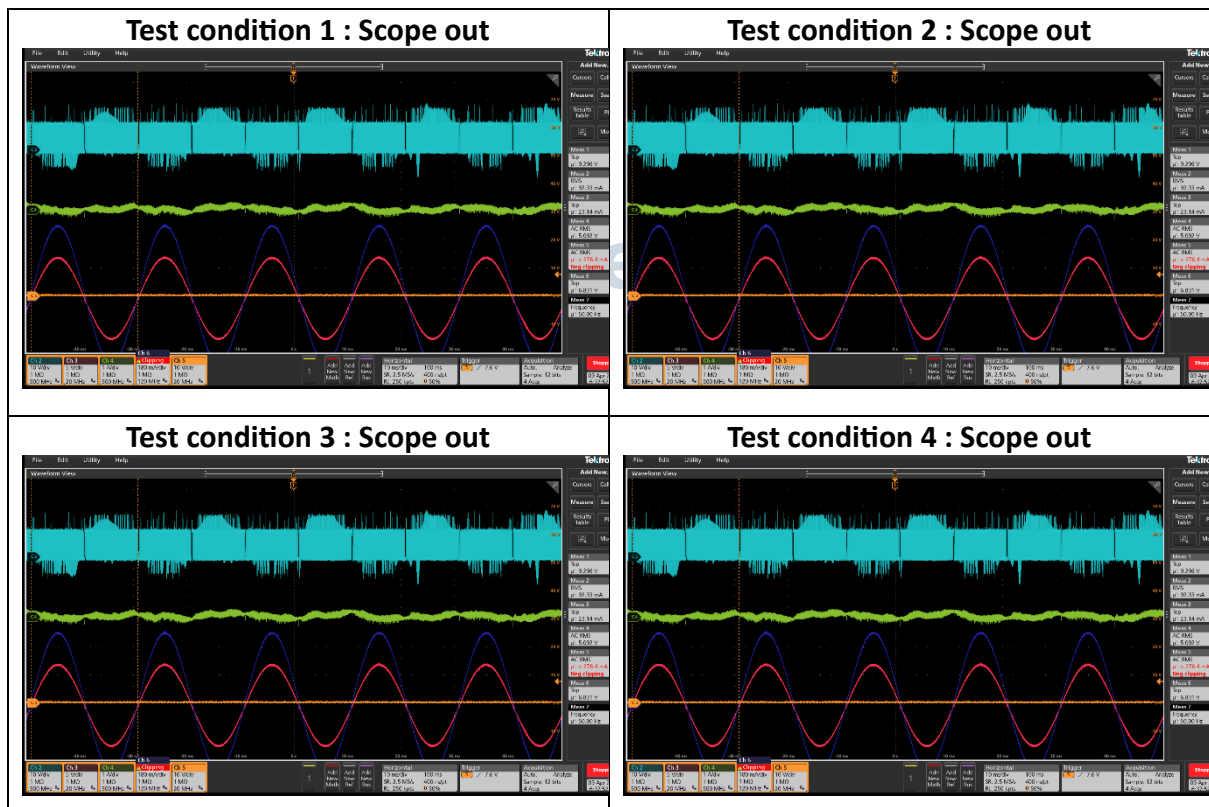
Timestamp: XXXXXXXX

Test Result: **Fail**

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



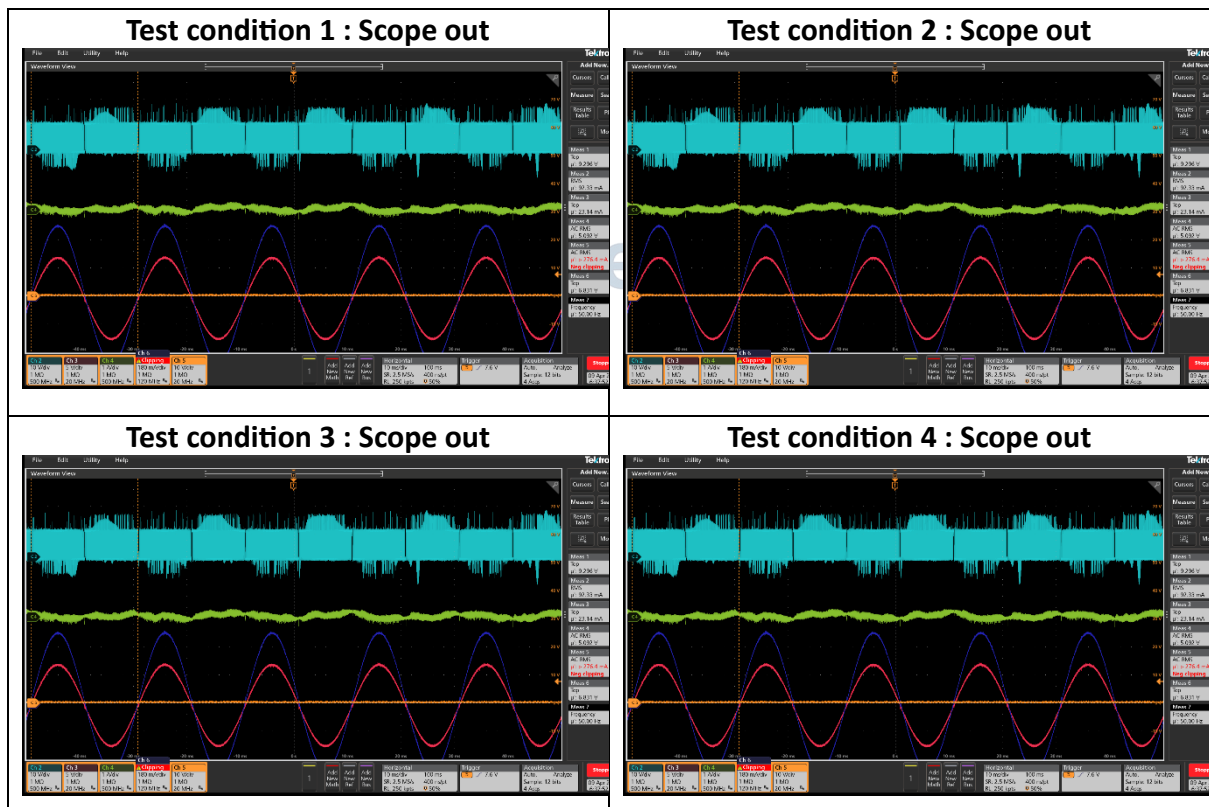
Timestamp: XXXXXXXX

Test Result: **Fail**

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



Timestamp: XXXXXXXX

Test Result: **Pass**

Test log

[10:36:30] Initializing Instruments...
[10:36:32] --- Starting Test: Input voltage range verification ---
[10:36:32] Running Global Setup (130% safety limits)...
[10:36:32] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[10:36:32] Setting Load Limits: I_{max}=23.85A
[10:36:32] Step 1: Setting Source Voltage to Nominal (80.0V)
[10:36:32] Step 2: Load Config -> Mode: CC, Value: 2.0
[10:36:32] Step 3: Turning Source ON
[10:36:32] Step 4: Waiting for V_{out} to reach 95% threshold (51.77V)...
[10:36:42] Timeout waiting for condition after 10s
[10:36:42] Error: Timeout waiting for V_{out} to reach 95% nominal
[10:36:42] --- Starting Test: Output voltage accuracy ---
[10:36:42] Running Global Setup (130% safety limits)...
[10:36:42] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[10:36:42] Setting Load Limits: I_{max}=23.85A
[10:36:42] Step 1: Setting Source Voltage to 48.0V
[10:36:42] Step 2: Turning Source ON
[10:36:43] Step 3: Measuring output voltage at no load
[10:36:43] No Load V_{out}: 24.063V
[10:36:43] Error: 55.85%
[10:36:43] Step 4: Testing at 25% load (4.59A)
[10:36:45] V_{out}: 23.925V, Error: 56.10%
[10:36:45] X FAIL: Exceeds 5.0% tolerance
[10:36:45] Step 5: Testing at 50% load (9.17A)
[10:36:47] V_{out}: 23.919V, Error: 56.11%
[10:36:47] X FAIL: Exceeds 5.0% tolerance
[10:36:47] Step 6: Testing at 75% load (13.76A)
[10:36:49] V_{out}: 24.065V, Error: 55.84%
[10:36:49] X FAIL: Exceeds 5.0% tolerance
[10:36:49] Step 7: Testing at 100% load (18.35A)
[10:36:51] V_{out}: 23.947V, Error: 56.06%
[10:36:51] X FAIL: Exceeds 5.0% tolerance
[10:36:51] X OVERALL FAIL: Output voltage accuracy out of specification
[10:36:52] --- Starting Test: Load regulation ---
[10:36:52] Running Global Setup (130% safety limits)...
[10:36:52] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[10:36:52] Setting Load Limits: I_{max}=23.85A
[10:36:52] Running Load regulation...